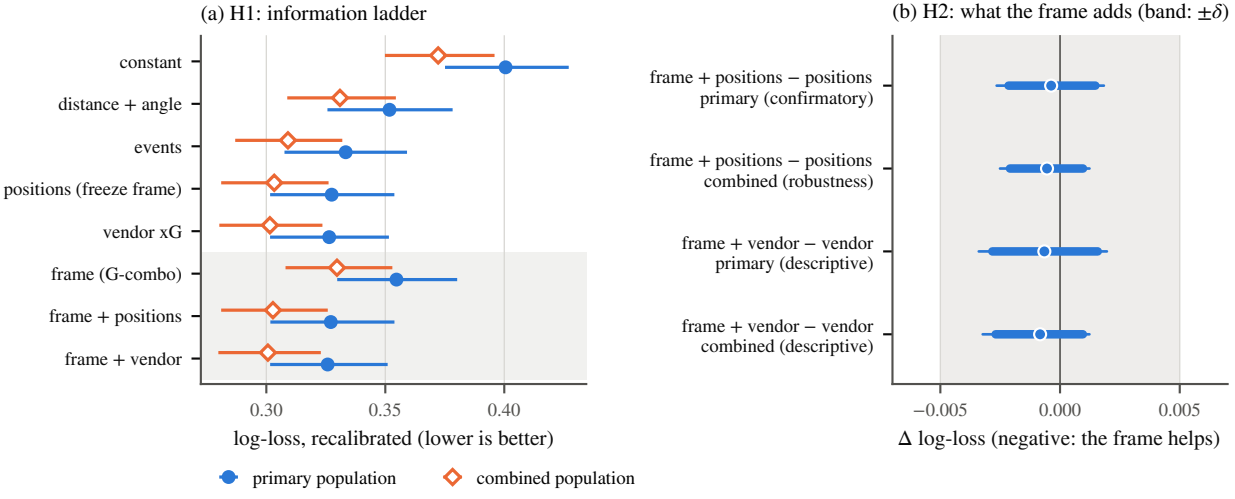
(a) H1: information ladder
constant
distance + angle
events
positions (freeze frame)
vendor xG
frame (G-combo)
frame + positions
frame + vendor
0.30
0.35
0.40
log-loss, recalibrated (lower is better)
primary population
combined population
(b) H2: what the frame adds (band: ±δ)
frame + positions − positions
primary (confirmatory)
frame + positions − positions
combined (robustness)
frame + vendor − vendor
primary (descriptive)
frame + vendor − vendor
combined (descriptive)
−0.005
0.000
0.005
Δ log-loss (negative: the frame helps)